

KLATEN, INDONESIA

Azza Sawungrana

Head of Data at Geosquare

National datasets, the standards around them, and what they cost to serve.

9 yrs	-32%	$23	4
GEOSPATIAL DATA	INFRASTRUCTURE COST	ALL NATIONAL DATA, MONTHLY	PEOPLE LED

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PROFILE

Head of Data at Geosquare, where national datasets become location intelligence for land, hazard, property and mobility decisions. I own the platform architecture, the standards the team works to, and what it costs to run.

The thread through Bhumi Varta and Geosquare has been the same one: taking delivery that was built for one client and making it work for the next. My work sits where data engineering meets product judgement, and I care about the cost of serving the second client, not only the quality of the first map.

EXPERIENCE

AUG 2025
PRESENT

Head of Data

Geosquare · Jakarta, Indonesia (remote)

Founding team since 2023, when the company was Petain. Head of Geospatial part-time 2024-25, full-time from August 2025.

A location intelligence company building grid-based data products for land, property, risk and network decisions.

- Lead the data function: platform architecture, engineering standards, the team, and what it costs to run.
- Took a platform built for one national land agency and sold it to two more clients, with a third in discussion. The first client is now on its third phase.
- Cut total cloud spend by 32%. The whole national dataset now sits in object storage for about $23 a month, so a new client adds compute rather than another copy of the data.
- Moved the bus factor from 2 people who could run the pipeline end to end to 5.
- Wrote the platform and portfolio strategy: six reusable engines, six product lines, and the sequence for getting from one to the other.
- Building KYL, a recurring product priced by the area a client needs.

PYTHON · DUCKDB · GEOPARQUET · PMTILES · CLOUD COST · PLATFORM STRATEGY

MAY 2025
JUL 2025

Geospatial Analyst

Van Lanschot Kempen · Amsterdam, Netherlands

Internship with the SDG Farmland Fund, on how a global farmland portfolio reports on sustainability and climate risk.

- Evaluated the fund's existing data systems and defined the use cases worth building first.
- Wrote the conceptual design for geospatial analysis, visualisation and reporting across the portfolio.

SPATIAL FINANCE · CLIMATE RISK · FARMLAND

APR 2020
FEB 2023

Lead Product Manager

Bhumi Varta Technology · Tangerang, Indonesia

LOKASI Intelligence, a location intelligence platform used by Indonesian banks, retailers and FMCG companies to decide where to expand and how to run their networks.

- Shipped version 2 in April 2021, a full redesign and refactor of the v1 codebase.
- Led LOKASI Intelligence Enterprise and turned client-specific deployments into modular capability inside the main product, instead of one-off forks.
- Ran the product team for most of this period with no CPO above it, so strategy, roadmap and prioritisation were mine.
- Built the delivery process for a small team: requirement gathering, PRD standards, agile, and a DevOps setup that worked without a dedicated ops person.

PRODUCT STRATEGY · ROADMAPPING · SDLC · REFACTOR

FEB 2023
JUN 2023

Product Consultant

Bhumi Varta Technology · Remote

Handover role before moving to the Netherlands.

FEB 2018
MAY 2019

Founder and Head of Product

Nusadventure · Klaten, Indonesia

A sustainable tourism venture built with Desa Wisata Jarum.

- Built and ran the product, operations and content side.
- Closed it after two years when conversion could not support the team.

MAR 2017
JAN 2018

Geospatial Data Analyst

Coordinating Ministry for Economic Affairs · Jakarta, Indonesia

One Map Policy, the national effort to reconcile conflicting land designations.

- Built and integrated thematic permit maps (mining, cultivation rights, location permits) for provinces across Sumatra and Nusa Tenggara.
- Worked with regional planning agencies, land offices and mining departments to trace overlapping designations back to their source.

EDUCATION

SEP 2023
JUL 2025

MSc Spatial Engineering

University of Twente, ITC · Enschede, Netherlands

A project-based programme about solving societal problems with spatial data. Funded by the ITC4SE scholarship.

- Thesis: migrant children's well-being and urban green and blue spaces in Enschede.
- Projects: drought and qanat water management in Mashhad; green-blue-grey space mapping for Kampala; local EV charging for a congested grid in Zwolle.

SEP 2012
FEB 2017

BSc Cartography and Remote Sensing

Universitas Gadjah Mada · Yogyakarta, Indonesia

Thesis on lahar (volcanic mudflow) modelling for Kelud, using field samples from three municipalities.

- Laboratory assistant for photogrammetry, disaster mapping and non-photographic remote sensing.
- Led the Geography Study Club for a year.

PUBLICATIONS

Geosquare System v1 - a nation-centric hierarchical grid
Chapter in a UGM Press volume edited by Trias Aditya · In press

Geosquare System v2 - country datums and per-country coverages
arXiv preprint · In preparation

Age-cohort dasymetric allocation to 50 m cells (RIUH)
FIT ISI 2026, Semarang · Presented; IOP proceedings in preparation

SKILLS

DATA ENGINEERING

Python, DuckDB, PostGIS, GDAL, GeoParquet, PMTiles, Cloud optimised GeoTIFF, object storage, cloud cost control

GEOSPATIAL

remote sensing, discrete global grid systems, dasymetric mapping, isochrones and routing, multi-criteria suitability, hazard modelling, spatial validation

METHODS

Ridge, random forest, gradient boosting and kernel models, spatial and temporal holdouts, uncertainty reporting, reproducible pipelines

LEADERSHIP

hiring, performance management, technical writing and standards, product strategy, cross-disciplinary delivery

LANGUAGES Indonesian (Native) · Javanese (Native) · English (Professional working proficiency)